CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 7: The Mathematics of Maybe: Introduction to Probability

Every formula and result of the chapter on one sheet

1	Range of a probability	$0 \le P(E) \le 1$
2	Impossible event	$P = 0$
3	Certain event	$P = 1$
4	Experimental probability	$\dfrac{\text{Number of times the event occurred}}{\text{Total number of trials}}$
5	Theoretical probability	$\dfrac{\text{Number of favourable outcomes}}{\text{Number of possible outcomes}}$
6	Complement	$P(\text{not } E) = 1 - P(E)$
7	Estimate for a population	Probability from the sample $\times$ size of the population
8	Sample space of one coin	$\{H, T\}$
9	Sample space of one die	$\{1, 2, 3, 4, 5, 6\}$
10	Two coins	$\{HH, HT, TH, TT\}$
11	Number of outcomes for two steps	(outcomes of step 1) $\times$ (outcomes of step 2)
12	Along a path of a tree diagram	Multiply the probabilities
13	Several paths	Add the probabilities
14	Law of Large Numbers	With more trials, experimental probability approaches theoretical probability